

Mohamed Amine Ben Zahra

Data Scientist & AI Engineer

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Rabat, Morocco

PROFESSIONAL SUMMARY

Data Scientist and AI Engineer operating at the intersection of data engineering, machine learning, and generative AI. Passionate about bridging complex data with practical applications that positively impact people's lives. Designs end-to-end ML pipelines, fine-tunes LLMs, and ships production-ready AI systems. Brings self-discipline, a collaborative mindset, and a dedication to continuous learning — key assets for driving success in distributed teams.

PROFESSIONAL EXPERIENCE

End-of-Year Intern — CDG Capital — Rabat, Morocco Jul. – Sept. 2025

- Designed an AI cognitive assessment prototype: comparative study of scoring performance, cognitive quiz generation, and a personalized recommendation RAG chatbot.
- Stack: LLM, RAG, Gemini API, REST API, LangChain, Embeddings, React.js, HuggingFace, MongoDB.

End-of-Year Intern — Web Help — Rabat, Morocco Jul. – Aug. 2024

- Designed an automated ticketing system with third-party API integrations and conversational agents.
- Stack: WordPress, PHP, Flask, Node.js, OpenAI API, SendGrid, Dialogflow.

PROJECTS

Diabetic Retinopathy Detection — Deep Learning Project May 2025 – Jun 2025

- Designed and deployed a high-performance medical diagnostic model with interpretability features, achieving a ROC-AUC of 0.94 and exposing RESTful endpoints via FastAPI.
- Implemented model versioning and experiment tracking using MLflow, and containerized the inference API for reliable deployment.

BERT Fine-Tuning — Sentiment Analysis Project May 2025

- Optimized sentiment analysis pipeline for CPU inference to under 80 ms through model quantization and parameter-efficient fine-tuning with LoRA/QLoRA.
- Built a low-latency inference service by exporting the fine-tuned DistilBERT model to ONNX and integrating it with a Python-based API.

Healthcare Visit Analytics — Business Intelligence & Data Engineering Apr 2025

- Built an end-to-end decision pipeline covering qualified data ingestion and patient monitoring dashboards.
- Stack: Power BI (DAX), dbt, Great Expectations, Python, SQL.

California Housing Regression — MLOps Project Nov 2024

- Implemented a complete MLOps workflow including model benchmarking, versioning, and continuous integration, achieving a MAE improvement of 18%.
- Stack: PyTorch, XGBoost, DVC, MLflow, Optuna, GitHub Actions.

EDUCATION

PhD 2024 – Present

Faculty of Sciences, Rabat (FSR-Rabat)

Master's, Data Science and Engineering 2022 – 2024

Faculty of Sciences, Rabat (FSR-Rabat)

Professional Licence, Networks and Telecommunications 2021 – 2022

Faculty of Sciences, Rabat (FSR-Rabat)

TECHNICAL SKILLS

Programming Languages	Python, Java, JavaScript, C++, PHP, C#, R
Frameworks	React, Django, Flask, .NET
Data Science & ML	Scikit-learn, TensorFlow, Keras, PyTorch, Pandas, NumPy, NLTK, SpaCy, ARIMA, LSTM, GRU, MLOps/AIOps, Statistics
NLP & GenAI	BERT, DistilBERT, LoRA/QLoRA, ONNX, Model Quantization, LLM Fine-tuning, Prompt Engineering
Data Engineering	PySpark, Kafka, dbt, Great Expectations, BigQuery, Power BI (DAX), ETL Pipelines
Web Scraping	BeautifulSoup, Selenium, Scrapy
APIs & Deployment	FastAPI, Flask, RESTful APIs, Docker, Kubernetes
DevOps & MLOps	Git, GitHub Actions, CI/CD, MLflow, DVC, Optuna, Container Orchestration, Log Monitoring
Databases	MySQL, PostgreSQL, SQL Server, SQLite, MongoDB, Oracle, BigQuery
Data Analytics	Cohort Analysis, User Segmentation, Product Analytics, Data Visualization
Methodologies	Agile, Scrum, JIRA, Planning, Resource Management

LANGUAGES

Arabic	Native
English	Bilingual
French	Bilingual
Spanish	Intermediate

CERTIFICATIONS

365 Data Science — Nov. 2025

✓ Introduction to Data & Data Science	✓ Machine Learning in Python
✓ Deep Learning with TensorFlow 2	✓ Understanding Cloud Computing
✓ The Complete Data Visualization Course with Python, R, Tableau	✓ Intro to AI Agents and Agentic AI
✓ Power BI	✓ SQL
✓ Build Chat Apps with OpenAI and LangChain	✓ Data Cleaning and Preprocessing
✓ Git and GitHub	

DataCamp

✓ Web Scraping in Python	✓ Introduction to Apache Kafka
✓ Introduction to Docker	✓ Introduction to Kubernetes
✓ Introduction to PySpark	

EXTRACURRICULAR ACTIVITIES

- Club Jeunes Leaders Marocains (JLM-FSR) (2022–2023) — Leadership initiatives.
- Club INNOVASUS-FSR (2024–2025) — Student innovation and entrepreneurship projects.

SOFT SKILLS

Problem Solving • Creativity • Critical Thinking • Time Management • Collaboration